Problem Challenge 1: Flip and Invert an Image (hard)

Problem Statement

Given a square binary matrix representing an image, we want to flip the image horizontally, then invert it.

To flip an image horizontally means that each row of the image is reversed. For example, flipping [0, 1, 1] horizontally results in [1, 1, 0].

To invert an image means that each 0 is replaced by 1, and each 1 is replaced by 0. For example, inverting [1, 1, 0] results in [0, 0, 1].

Example 1:

```
Input: [
  [1,0,1],
  [1,1,1],
  [0,1,1]
]
Output: [
  [0,1,0],
  [0,0,0],
  [0,0,1]
]
```

Explanation: First reverse each row: [[1,0,1],[1,1,1],[1,1,0]]. Then, invert the image: [[0,1,0],[0,0,0],[0,0,1]]

Example 2:

```
Input: [
  [1,1,0,0],
  [1,0,0,1],
  [0,1,1,1],
  [1,0,1,0]
]
Output: [
  [1,1,0,0],
  [0,1,1,0],
  [0,0,0,1],
  [1,0,1,0]
]
```

Explanation: First reverse each row: [[0,0,1,1],[1,0,0,1],[1,1,1,0],[0,1,0,1]]. Then invert the image: [[1,1,0,0],[0,1,1,0],[0,0,0,1],[1,0,1,0]]

Try it yourself

Try solving this question here:

Java

```
class Solution {
  public int[][] flipAndInvertImage(int[][] arr) {
    // TODO: Write your code here
    return arr;
  }
}
```

Test Case | Run Code Result | Stdout

[[1,0,1], [1,1,1], [0,1,1]]

Use Example Testcases | Run | Submit